SMOOTH CENTRALLY SYMMETRIC CONVEX DOMAINS IN HIGH DIMENSIONS WITH INTERIOR HOT SPOTS

ARGUS OPEN RESEARCH PROJECT

Abstract. We document the counterexample of de Dios Pont (2024) to the hot-spots conjecture for centrally symmetric smooth convex domains in $\mathbb{R}^d$ for sufficiently high dimension d.

1. Main Refutation

Theorem 1.1 (de Dios Pont, 2024)**.** *For all sufficiently large dimensions d, there exists a smooth, bounded, centrally symmetric convex domain $\Omega \subset \mathbb{R}^d$ such that every second Neumann eigenfunction u of $-\Delta$ on Ω attains its strict global maximum and minimum in the interior of Ω rather than on the boundary $\partial\Omega$.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 20000496 in the domain of spectral geometry and partial differential equations follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *published refutation* deciding the scope: hot-spots conjecture for centrally symmetric smooth convex domains in sufficiently high dimensions (de Dios Pont 2024).

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- low-dimensional planar/3D centrally symmetric convex domains.

References

[1] Primary problem list and source literature for Result 20000496 (spectral geometry and partial differential equations), archived with exact theorem locators and verification certificates in the accompanying repository package.

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